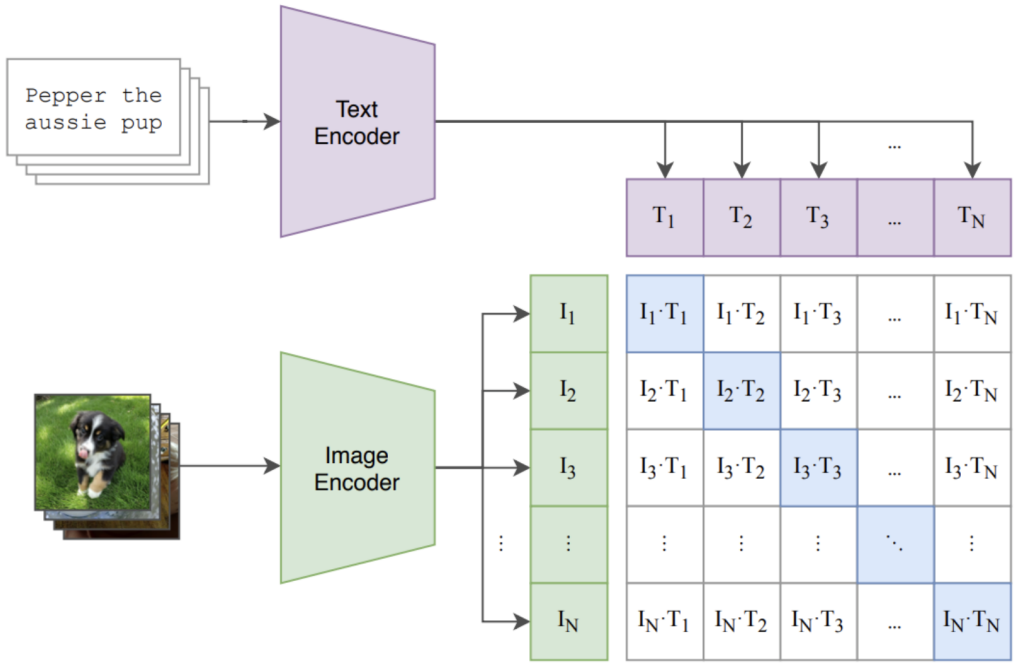
Contrast Language Image Pretraining



1. **Logic Chain:**

   This model aligns images and texts semantic meanings in the shared space.

   It encodes the images with ViT or ResNet, then encodes the texts with BERT-like encoder.

   The the CLS tokens from ViT or pooled features from ResNet, together with text embeddings are used in contrastive learning.

2. **Training Process:**

   Take CLS as an example, the model uses the matrix shown in the image to calculate the similarities for all the possible pairs.

   Then the target is to minimize the InfoNCE:

$$L = \frac{1}{2}L_{t\rightarrow i} + \frac{1}{2}L_{i\rightarrow t}$$

   For InfoNCE, the loss is

$$L_{InfoNCE} = \frac{exp(sim^{+})/\tau}{\sum_i exp(sim)/\tau}$$

   So when we are minimizing the InfoNCE, we are making positive pairs closer and negative pairs further apart at the same time.

3. **Embeddings Normalization:**

Before calculating cos similarity, embeddings are normalized to unit length.

The reason is the lengths should not contain semantic meanings.

It is because during the transformer operations, embeddings are usually randomly stretched, compressed, making the length too noisy to carry semantic meaning.

4. **Why bidirection loss:**

   Because we are summing over the different dimension of the matrix in different direction.

   In $i \to t$, we fix the image, and sum over all the texts, which is the dim=1 in the shown matrix.

   While in $t \to i$, we are summing over the dim=0.

5. **ViT VS ResNet:**

   Usually ViT is preferred, because of the following reasons.

   The ResNet is based on CNN, which has a high bias, which is the pixels are sharing the same kernel.

   The good part of it is it requires less data to train the model, the downside is it limits the model's expression ability. With flexible self-attention, ViT can capture semantic meanings better.

   And if we want to increase the depth, we can simply stack encoders in ViT, but for ResNet, we should carefully tune designs like kernel size, stride, padding, which is inconvenient.

   ViT is often already pretrained well, with a good zero-shot accuracy, and the output is transformer embeddings, which are genuinely suitable for the shared space.

   But ResNet has its own advantage.

   It usually requires less data because in each layer the kernels are shared.

   And when the resolution is increased, the burden for ViT increases much faster than ResNet.

6. **CLIP usage:**

   CLIP is not used for generation of things.

   It is usually used in scoring based actions, like retrieval and evaluation.

7. **Dense Alignment:**

   In contrastive learning, the similarity between [CLS] and text embeddings are optimized.

   Even though the update can flow from [CLS] to image dense embeddings, it is not guaranteed that the dense embeddings are accurate enough, because at least there isn't a loss involving them.

   So there are methods using dense embeddings directly instead of [CLS] for contrastive learning, making the dense representations more accurate.